

# DESIGNING RFID IMPLANTS

HOW FLIPPING THE BIRD OPENS DOORS FOR ME

**WHAT AM I TALKING ABOUT?**

THERE WILL BE NO  
GORY IMAGES



**TARGET AUDIENCE?**

**WHO AM I?**

# MIANA WINDALL

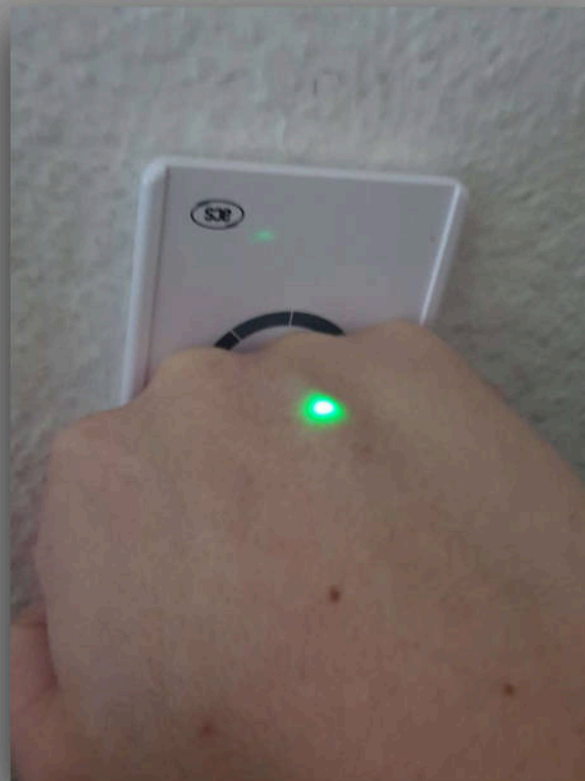
**SHE/HER**

COMPUTER NERD BY DAY

BIOHACKER BY NIGHT



10 RFID,  
8 MAGNET,  
2 AESTHETIC,  
6 MEDICAL



xSIID NFC Implant

**LET'S GET  
STARTED**





# WHAT IS RFID?

# KEY TERMINOLOGY

**RFID** RADIO FREQUENCY IDENTIFICATION

**LF** LOW FREQUENCY (125KHz / 132KHz)

**HF** HIGH FREQUENCY (13.56MHz)

**NFC** STANDARDIZED SUBSET OF HF RFID

**TAG** PASSIVE RFID TRANSPONDER

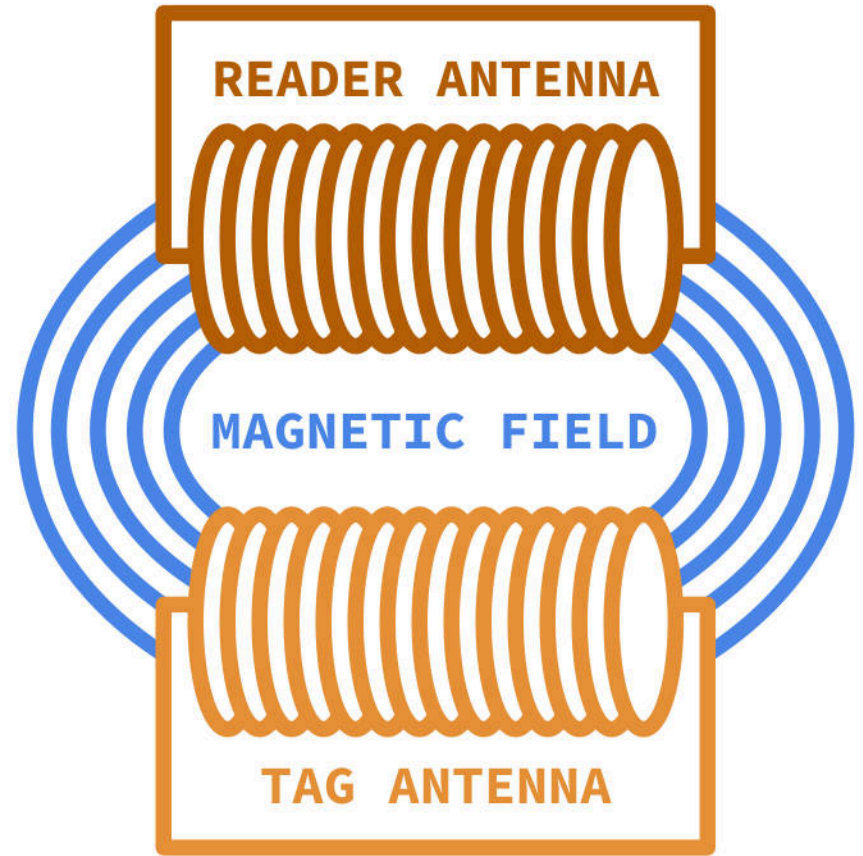
**READER** ACTIVE DEVICE TALKING TO TAG

**FIELD** THE EM FIELD OF A READER & TAG

# HOW DOES IT WORK?



# HOW DOES IT WORK?

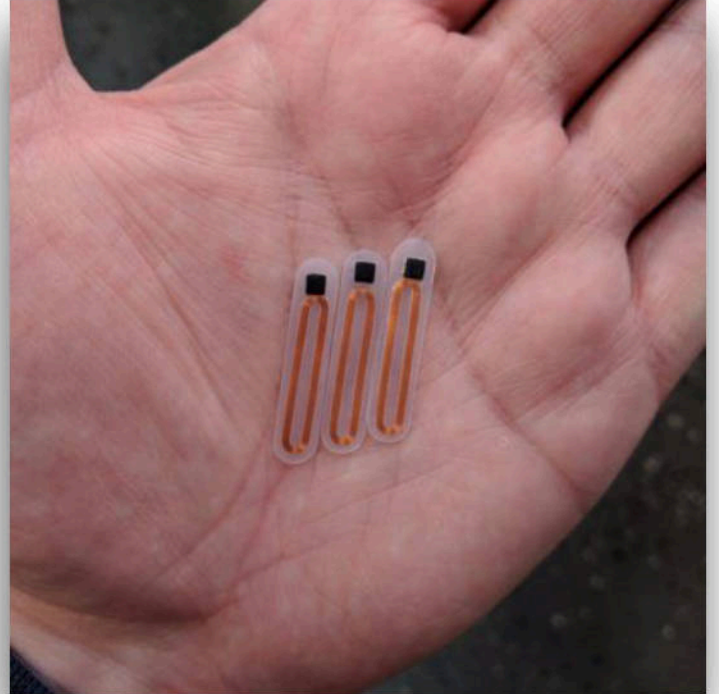


**WHAT DO I MEAN BY IMPLANT?**

# SO WHAT IS AN IMPLANT?



# SO WHAT IS AN IMPLANT?



# SO WHAT IS AN IMPLANT?



X-RAY BY CASEY BORDERS



# SO WHAT IS AN IMPLANT?



X-RAY BY CASEY BORDERS

# JUST RFID?

MAGNETS

OTHER INDUCTIVE POWER TECH

SILICONE

TRIDIUM

MEDICATION

ETC



TITAN SENSING MAGNET IMPLANTS





XGLO TRIDIUM IMPLANT - PHOTO BY ALYX WIJERS



POST INSTALL SILICONE POWER BUTTON SUBDERMAL

# JUST RFID?

MAGNETS

OTHER INDUCTIVE POWER TECH

SILICONE

TRIDIUM

MEDICAL

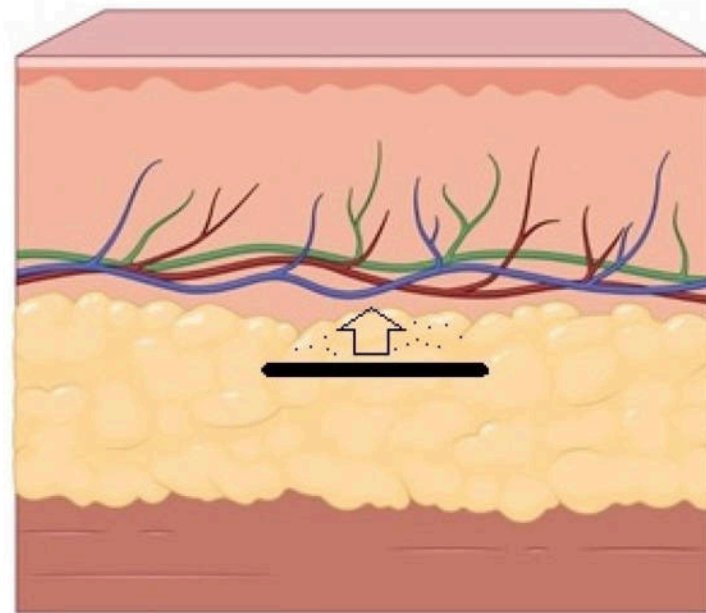
ETC

# GETTING IMPLANTED!

Epidermis  
Dermis

Subcutis

Muscle



**CONTENT WARNING:  
NEEDLES**







VARIOUS IMPLANT NEEDLES - PHOTO BY "JIRVIN" FROM THE DANGEROUS DISCORD



PRE INSTALL SILICONE SUBDERMAL – PHOTO BY NICK “TCPRST” DRAFFEN

# DISADVANTAGES?





# DISADVANTAGES?



**DISADVANTAGES?**



# DISADVANTAGES?



# ADVANTAGES?



ADVANTAGES?





ADVANTAGES?



# DESIGNING MY OWN RFID IMPLANT

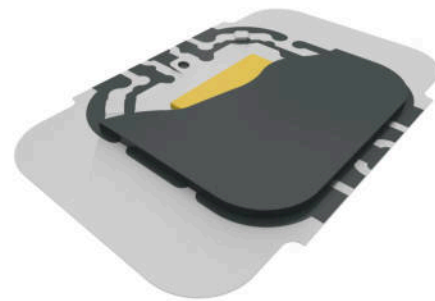
# WHY?

WHAT INSPIRED ME TO DO  
THIS?



# KEY ELEMENTS OF RFID TRANSPONDERS

# CHIP



# ANTENNA





# HOW DO THEY WORK TOGETHER?



LC Tank Circuit

# SOURCING THE PARTS

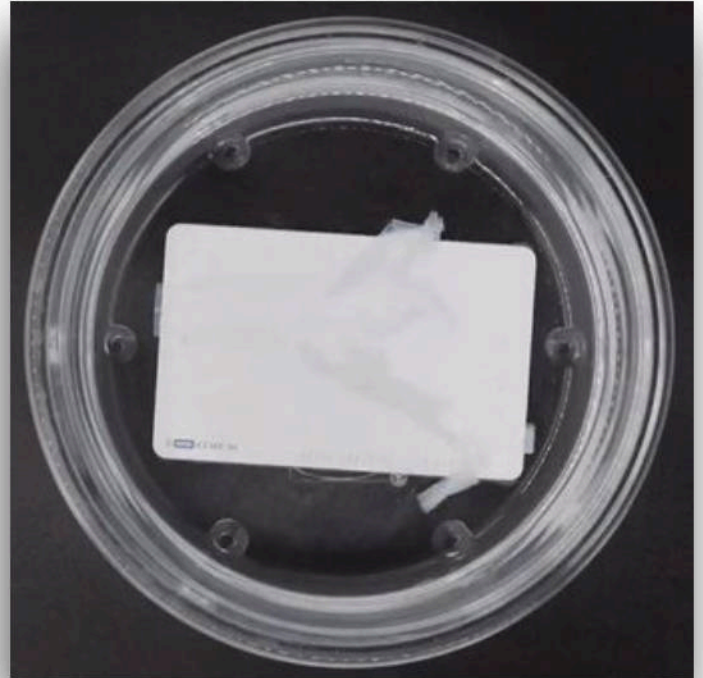
# PROPRIETARY CHIPS

ARE A PAIN IN THE ASS



# CHIP EXTRACTION

“I’M MELTING”



VIDEO BY NICK “TCPRST” DRAFFEN  
[doitfor.science/melting](https://doitfor.science/melting)

# DATA GATHERING

MEASURING THE CHIP  
CHARACTERISTICS



PHOTO BY NICK “TCPRST” DRAFFEN

**AFTER ALL THAT...**

**I FOUND THE DATASHEET...**

# DESIGNING THE IMPLANT



# DOING THE MATH

## LC Resonant Frequency Calculator

Inductance, L

50

$\mu\text{H}$  ▾

Capacitance, C

?

pF ▾

Resonant Frequency, f

13.56

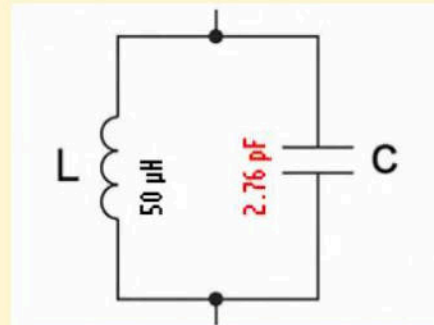
MHz ▾

Calculate

Reset

## Results

Capacitance (C) = **2.755185727 pF**



# DOING THE MATH

All units are in millimeters

Is the inductor round[0] or square[1]?

1

You selected square.

Enter the width[a0] of the NFC antenna:

23

Enter the height[b0] of the NFC antenna:

4.6

Enter the track thickness[t] of the NFC antenna:

0.03556

Enter the track width[w] of the NFC antenna:

0.1

Enter the track spacing[g] of the NFC antenna:

0.1

Enter the number of turns[Na] of the NFC antenna:

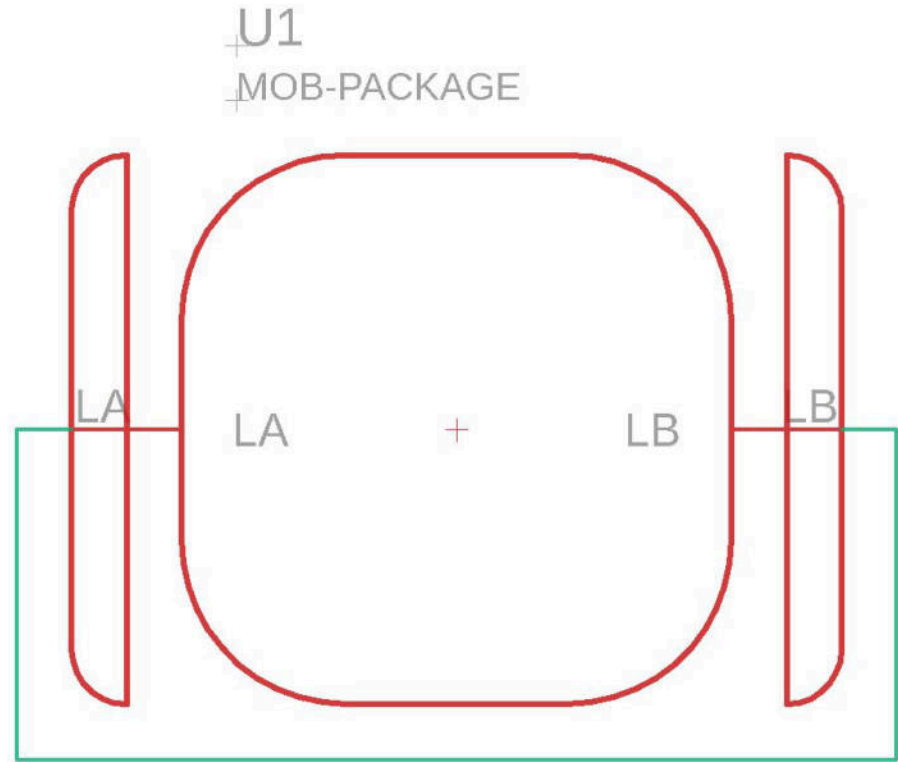
14

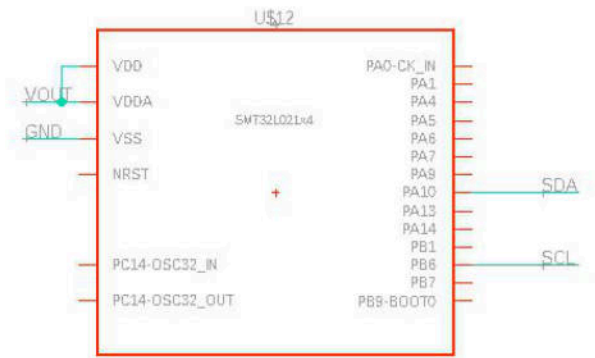
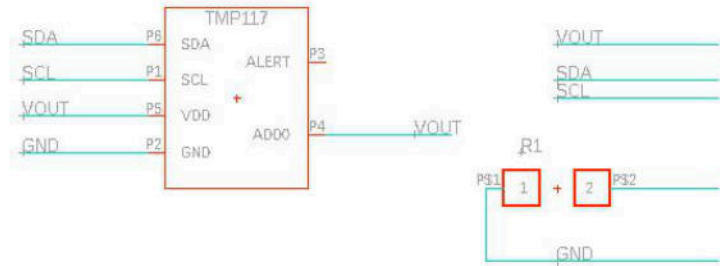
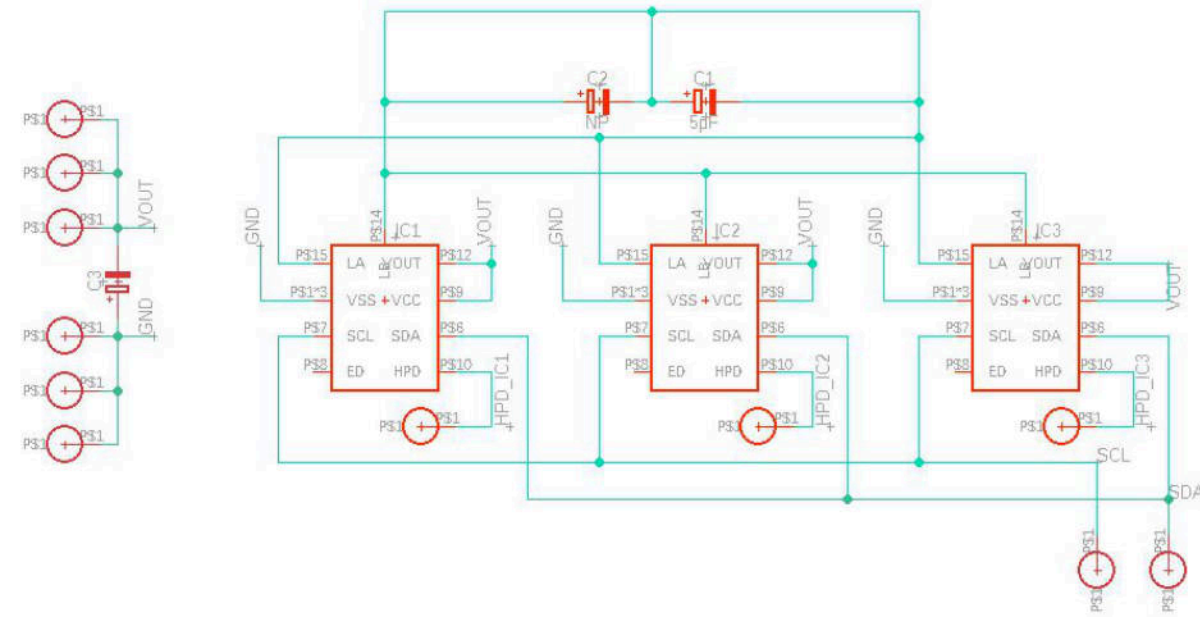
The inductance of the coil is:

3.234 $\mu$ H

**AUTOMATING THIS IS THE  
DREAM...**

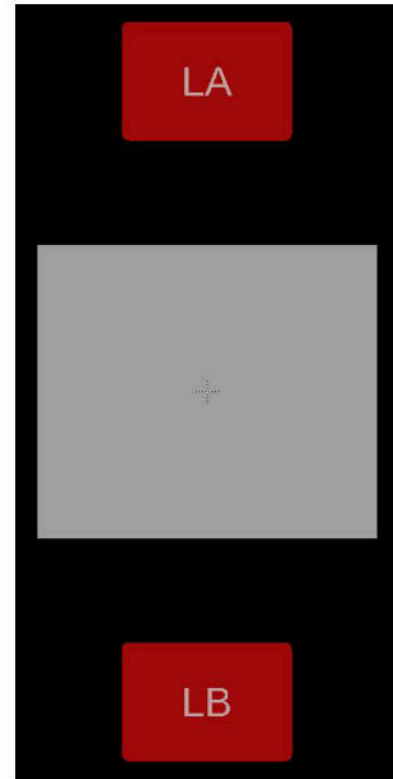
# CIRCUIT SCHEMATIC



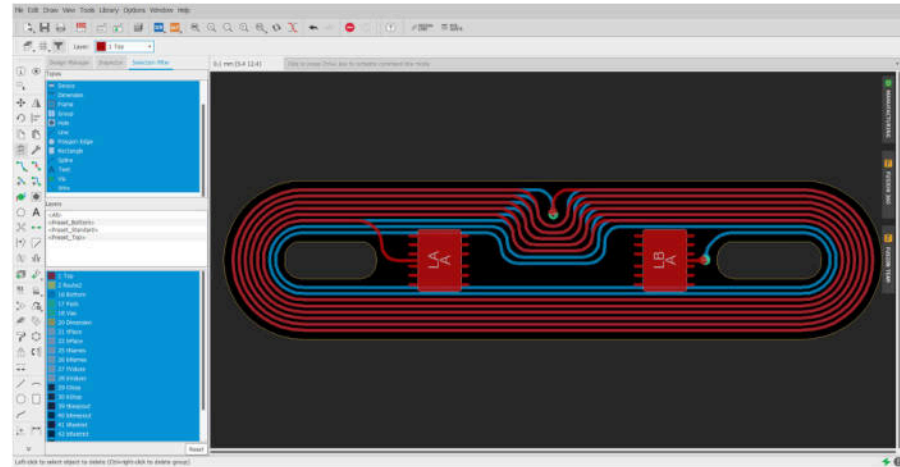


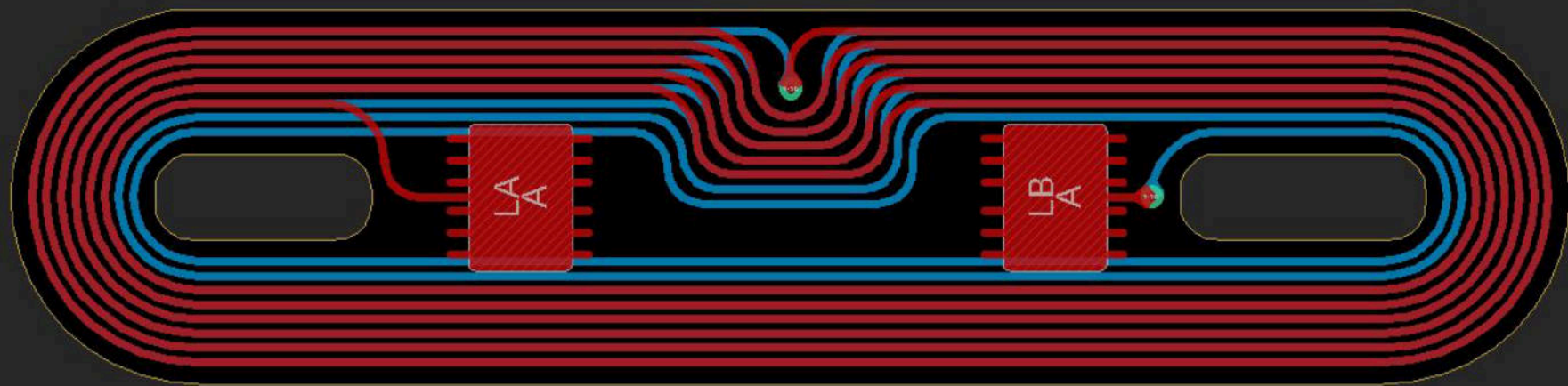
LITAF FLEX LIGHT PROTOTYPE SCHEMATIC

# FOOTPRINT DESIGN



# PCB LAYOUT

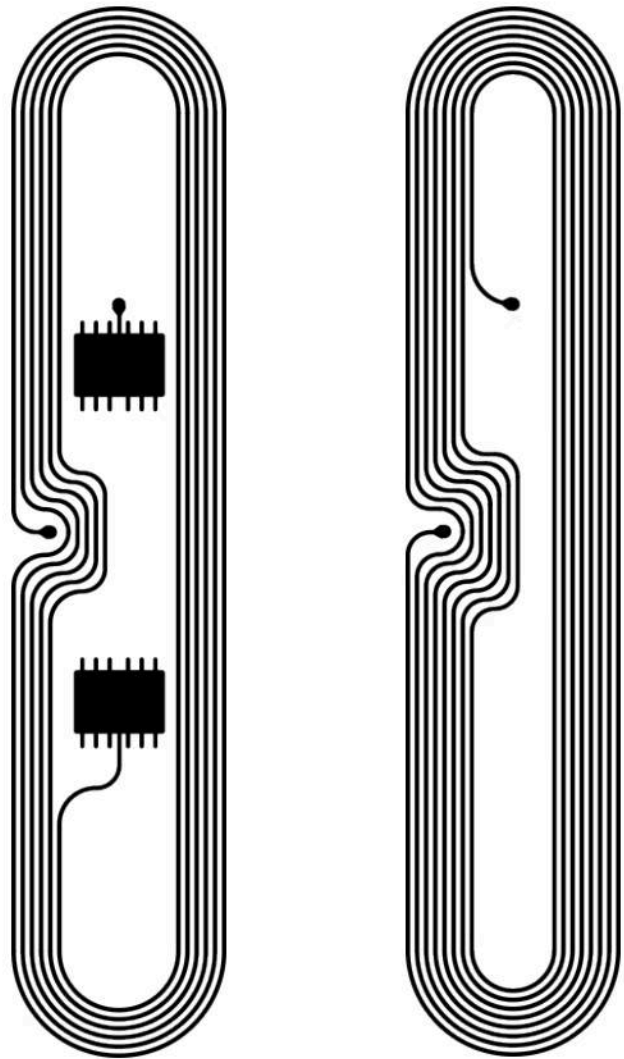




FLEXCLASS IMPLANT PCB LAYOUT



# MANUFACTURING



# MANUFACTURING

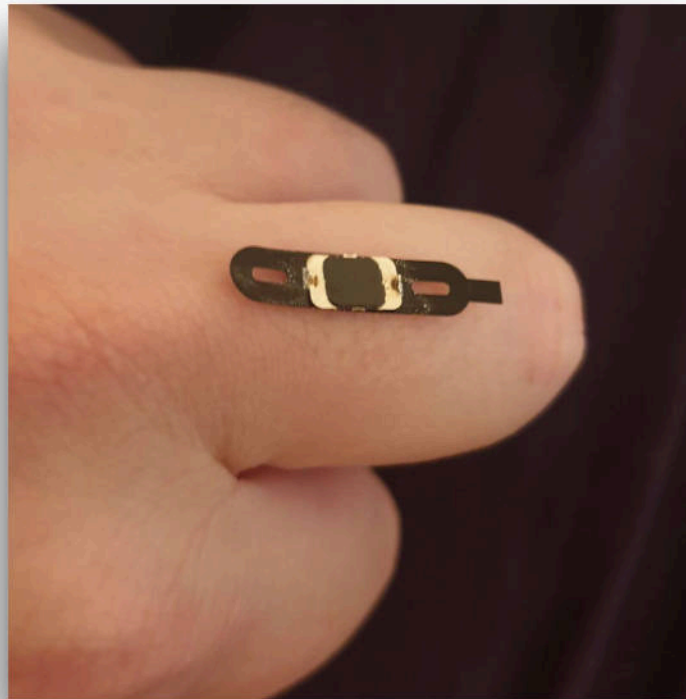


**THIS CAN BE A BIT COSTLY...**

# ASSEMBLY



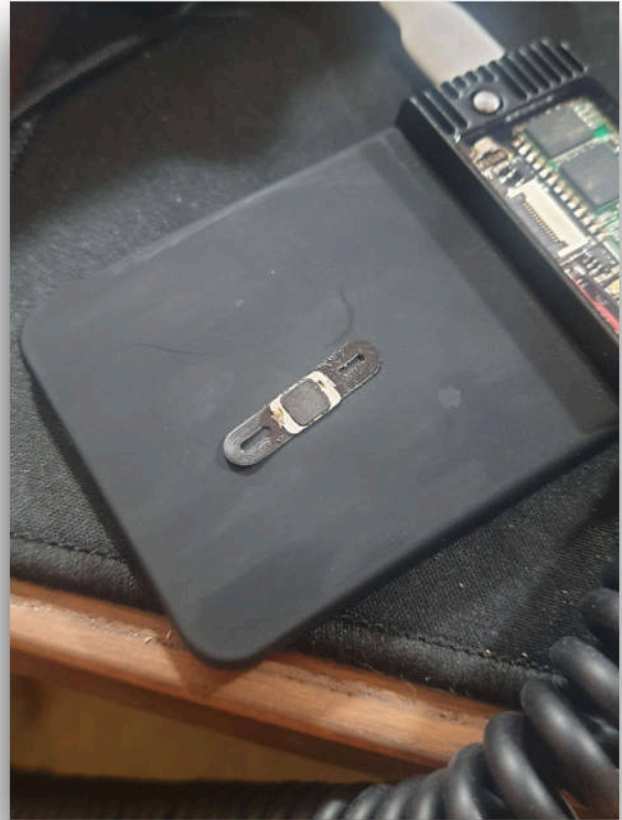
# ASSEMBLY



# TRIAL AND ERROR

# TESTING!

DOES IT WORK? HOW WELL?



# WHAT'S WRONG?

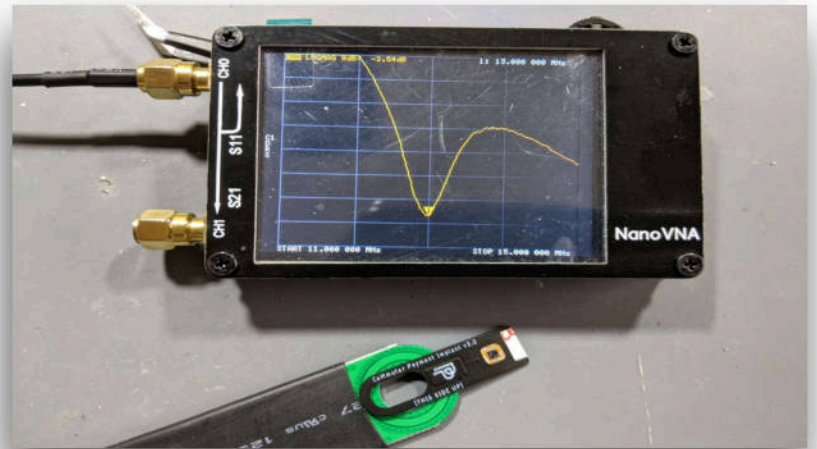


PHOTO BY CORY LORD



# HOW TO FIX IT?

## LC Resonant Frequency Calculator

Inductance,  $L$

50

$\mu\text{H}$  ▾

Capacitance,  $C$

?

$\text{pF}$  ▾

Resonant Frequency,  $f$

13.56

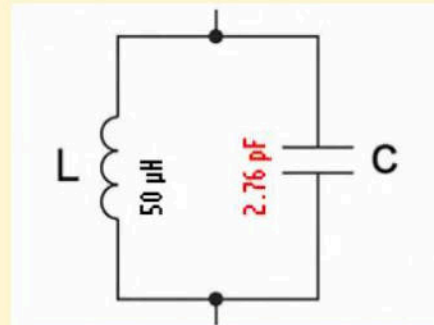
$\text{MHz}$  ▾

Calculate

Reset

## Results

Capacitance ( $C$ ) = **2.755185727  $\text{pF}$**



# PROCESS IMPROVEMENTS



PHOTO BY “🐝” FROM THE DANGEROUS DISCORD

**MAKING IT BIOCOMPATIBLE**

**THIS STEP IS ONLY NEEDED FOR  
IMPLANTS**

# BODY SAFE COATINGS

I LEAVE THIS TO THE  
PROFESSIONALS



**AFTER COATING**



# INSTALLATION

STABY STABY STAB



**SO WHAT WHERE THE RESULTS?**



# BOOM!

IT WORKS



[doitfor.science/bird](https://doitfor.science/bird)

**YOU MIGHT WANT TO ASK  
BEFORE YOU DO THIS...**

WHAT'S IT LIKE TO  
ACTUALLY USE IT?



[doitfor.science/bird](https://doitfor.science/bird)

# LESSONS LEARNED

WHAT IMPROVEMENTS WOULD I  
MAKE?



**JUST DO IT!**

**“THE FOOL DID NOT KNOW IT  
WAS IMPOSSIBLE SO SHE DID  
IT ANYWAY”**

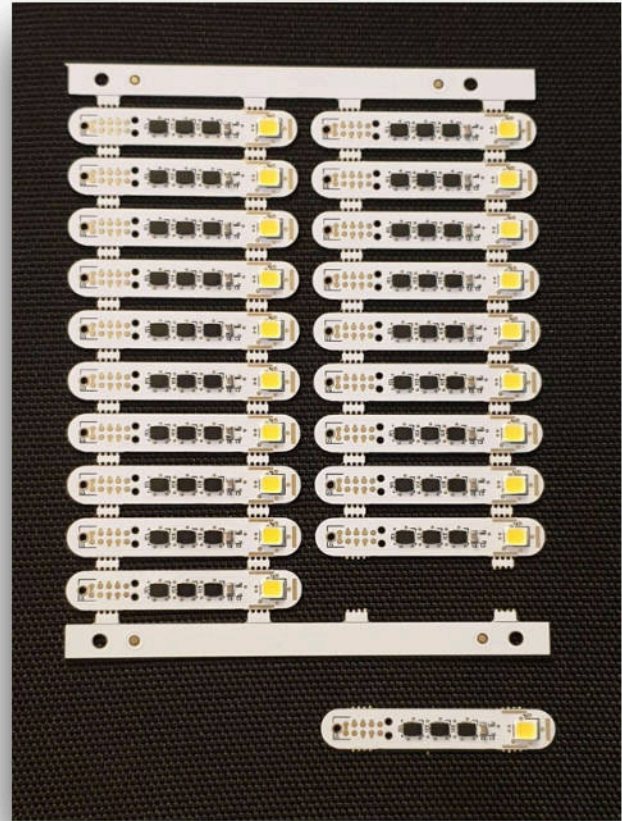
**WHAT IS NEXT?**

# WORK IN PROGRESS PROJECTS



# LITAF FLEX RGB

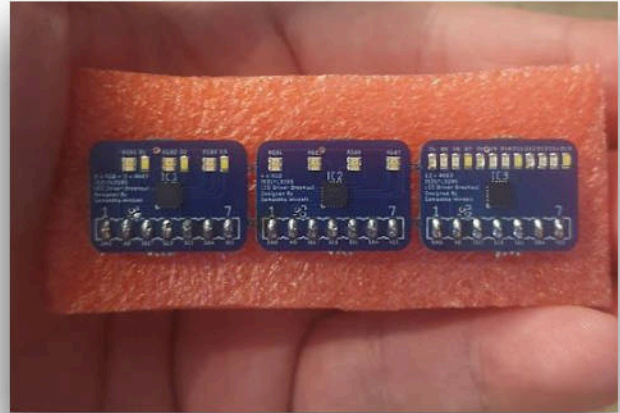
NFC POWERED ANIMATED RGB  
LED IMPLANT



**POWER, COMS, AND A MICRO...**

# LITAF FLEX RGB

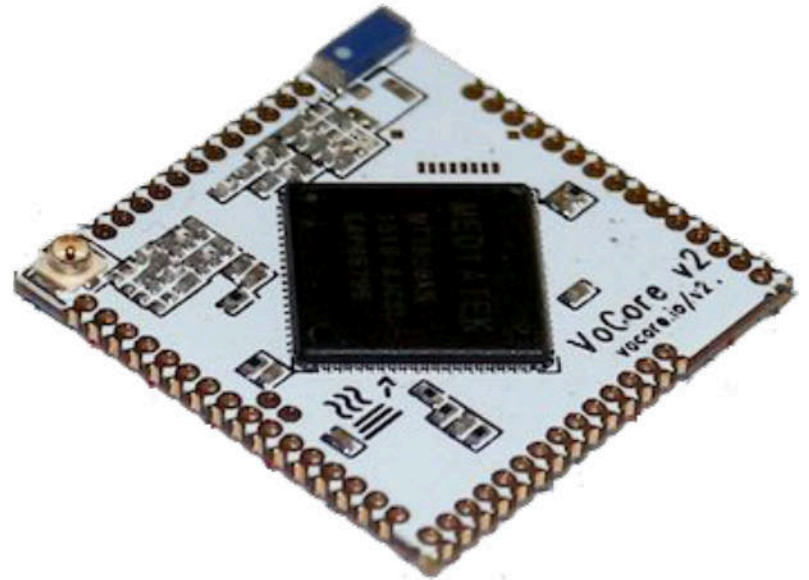
NFC POWERED ANIMATED RGB  
LED IMPLANT



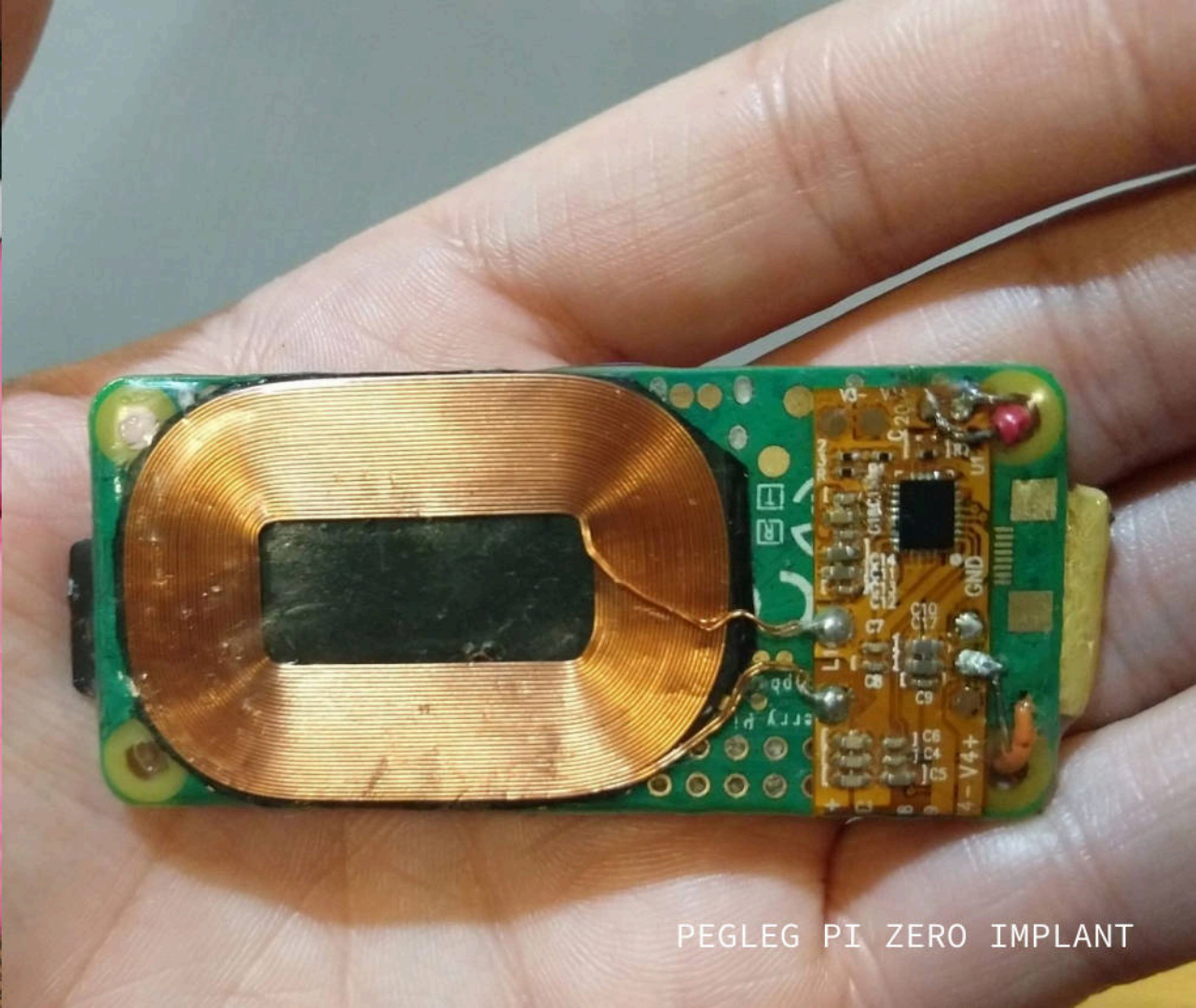
LitAF Platform V0

# BODYBYTES

QI POWERED WIFI ENABLED  
LINUX COMPUTER IMPLANT



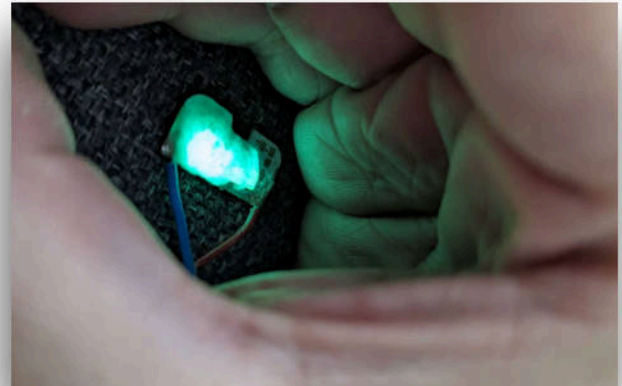
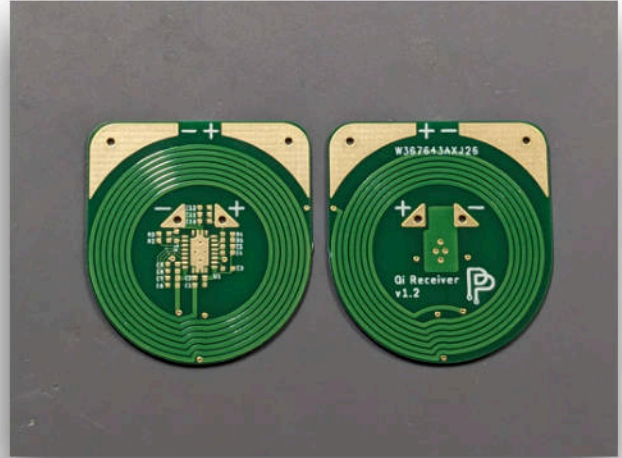




PEGLEG PI ZERO IMPLANT

# LITAF QILIGHT

A QI POWERED LED IMPLANT



# LITAF QILIGHT

A QI POWERED LED IMPLANT

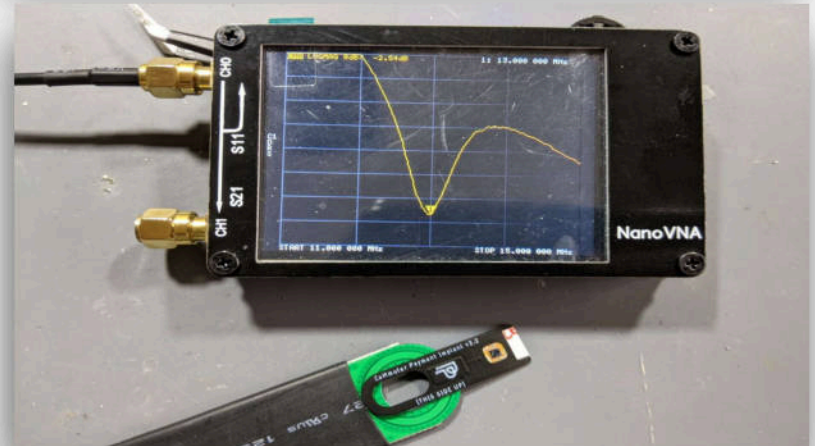


Tattoo by Maria Miller



# COMMUTER

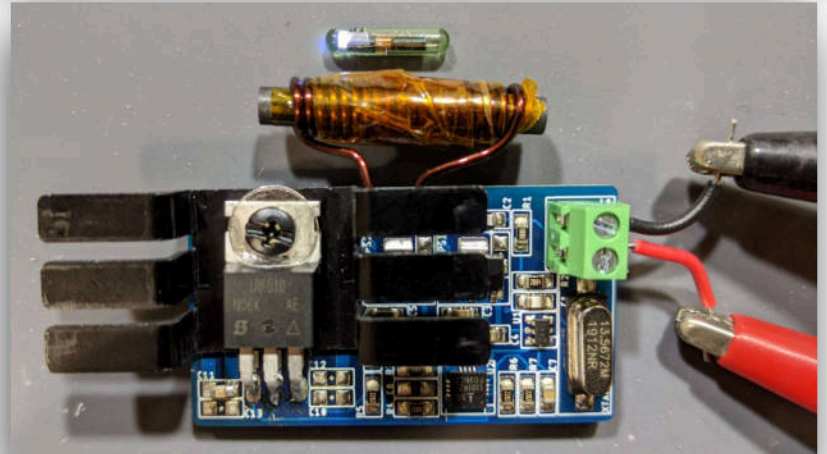
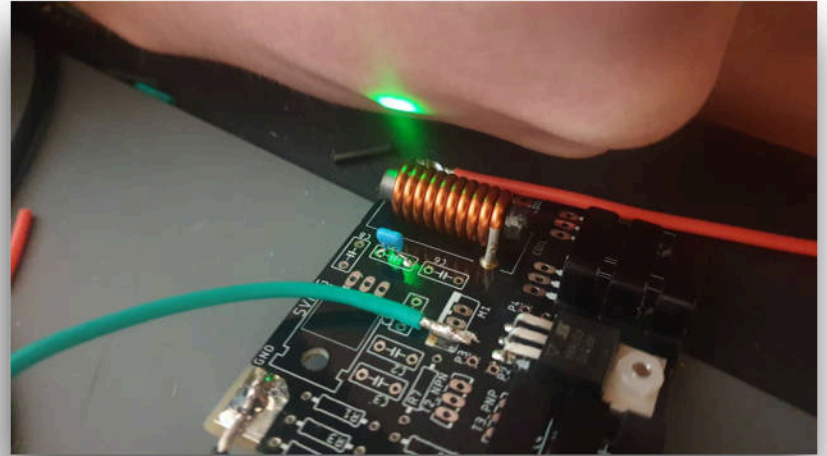
IMPLANT CONVERSION BOARD  
FOR COM BASED CREDIT CARD





# POWER ACCESSORIES

HF FIELD GENERATOR



# GET INVOLVED!

IF YOU WANT TO LEARN MORE  
COME JOIN US IN OUR  
DISCORD COMMUNITY



[doitfor.science/dd](https://doitfor.science/dd)

**FUTURE IMPLICATIONS OF THIS TECH?**

IS IT JUST A  
GIMMICK?



# WHERE IS IT GOING?

WHAT MIGHT THE FUTURE HOLD?



# WHAT ARE THE BARRIERS?



# WHAT ARE THE BARRIERS?



# THANK YOU!

The Biohacking Community

Cory “Satur9” Lord

Daniel “DonFire” O'Neill

The RFID Hacking Community

Nick “tcprst” Draffen

Dangerous Things

Johnross Switz

The DEF CON Goons & Community



# CONTACT ME

HAVE A QUESTION?  
WANT TO CHAT?  
GET IN TOUCH!



[doitfor.science/miana](https://doitfor.science/miana)

# RESOURCES

LINKS TO MY THESE SLIDES  
AND ANY OTHER RELEVANT  
RESOURCES



[doitfor.science/dc31/resources](https://doitfor.science/dc31/resources)

**QUESTIONS?**

# GOODBYE!

LINKS TO MY THESE SLIDES  
AND ANY OTHER RELEVANT  
RESOURCES



[doitfor.science/dc31/resources](https://doitfor.science/dc31/resources)